Biodiversity Decline

The decline of bird populations is an environmental concern because of their vitality to ecosystems and their indication of overall environmental health. Rosenberg et al. (2019) estimate a loss of 3 billion birds from 1970 to present. The main reason for this decline is habitat loss, but birds are also negatively affected by pollutants. Specifically, birds around the San Francisco Bay are negatively affected by the mercury concentration in fish in the bay (Luengen, 2026). Data collection for birds is easier to come by than other species due to recreational birders who are fascinated by the beautiful avifauna and document birds they see. In order to learn more about bird populations around San Francisco Bay, our class took a field trip to Aquatic Park Cove, led by one important birder, Corny Foster. This paper explores the decline of North American avifauna, the effect of pollutants on birds, and our observations at the San Francisco Aquatic Park Cove.

Decline of North American Avifauna

Rosenberg et al. (2019) emphasize that focusing only on extinction of species underestimates human impact. In order to truly understand the effects on our ecosystems, we must quantify the decline in populations. They observed population change for 529 species of birds in the continental US and Canada. They used 143 weather radars to estimate changes in nocturnal migrating patterns, estimating a net loss in total abundance of 2.9 billion since 1970. In order to better understand specific bird population decline, Rosenberg et al. (2019) observed the changes in biomass passage of birds specific to regions in the continental United States.

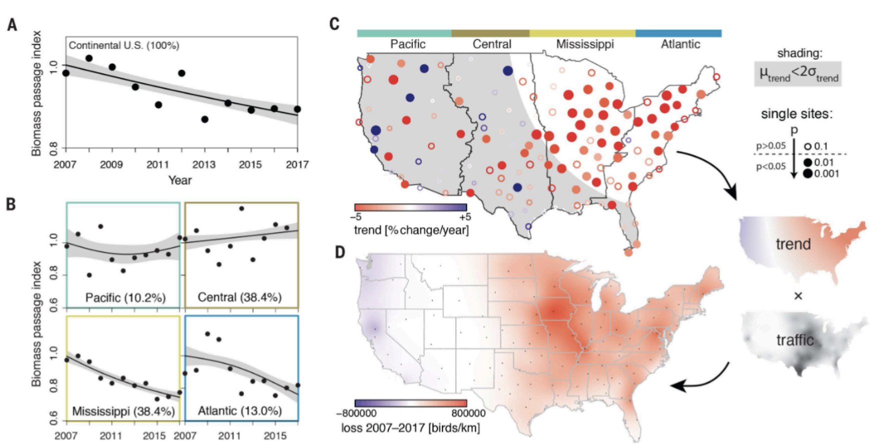

Figure 1 (from Rosenberg et al., 2019) *Long-term changes in nocturnal migration patterns, specifically observed per region of the continental United States.*

They gathered data using 143 NEXRAD stations. **Figure 1A** shows the annual change in biomass across the entire US, showing an overall decrease in biomass passage, a reduction of 13.6 ± 9.1% from 2007–2017. **Figure 1B** visualizes the change in biomass passage split up by region (Pacific, Central, Mississippi, and Atlantic), demonstrating that Mississippi and Atlantic regional biomass passage decreased over the 10 years while Pacific and Central slightly increased. **Figure 1C** visualizes the single-site trends of biomass passage measured at each NEXRAD station. Statistically significant changes were shown with filled-in circles, red circles indicating a significant decrease and blue circles indicating a significant increase. **Figure 1D** visualizes the ten-year cumulative loss in biomass passage as the product of a spatially explicit trend times the surface area cumulative spring biomass passage (Rosenberg et al., 2019).

Figure 1 demonstrates that the Mississippi and Atlantic regional birds suffer more from population decline than birds in the Pacific and Central areas, highlighting that population decline is a result of habitat loss. The Mississippi region is a crucial agricultural region dominated by crop production, leaving little room for bird habitats. The Atlantic region of the US is dense in population, also leaving little room for bird habitats. The Pacific region of the US is more spread out, leaving areas for birds to populate. The Pacific region also has conservation areas designated for bird populations to thrive.

Rosenberg et al. (2019) emphasize that their observations are an underestimate of the total bird population decline because they only observed migratory birds. Although they observed an increase in certain populations, specifically Pacific region birds, these increases are not able to counteract the losses. While they only observed bird populations, birds are an indicator of overall ecosystem health and suggest population decline in other species. Their results show an urgent need for conservation action and legislation in order to limit anthropogenic biodiversity loss.

Negative Effect of Pollutants on Birds

Although habitat loss is the main contributing factor of bird population decline, birds are also negatively affected by anthropogenic pollution. The biggest pollution concern for San Francisco birds is methylmercury. Fish that live in the bay demonstrate the high concentrations of methylmercury in the water. Striped bass, bat rays, and 4 additional species exceed the standard of 0.2 ppm. Methylmercury biomagnifies, meaning it becomes more concentrated at higher trophic levels of the food chain (Luengen, 2026). Therefore, mercury is a high concern for subsistence fishers who feed their families fish from the bay, as well as birds who are consuming this fish.

The effects of mercury on birds are widely studied. It has been found that bird body condition is negatively correlated with mercury concentration, and MeHg has been found to decrease adult survival and impair behavior. Bird populations are an indicator of overall ecosystem health, and the negative effect of mercury on San Francisco Bay birds emphasizes the need for mercury concentration monitoring in the bay in order to reduce pollution and improve overall ecosystem health (Luengen, 2026).

Aquatic Park Cove Field Trip

Originally, our class was supposed to go to Alcatraz to bird watch, but due to dock repairs we were unable to. Alcatraz is a good example of seabird habitat. Since the US Army fortified Alcatraz in 1859, it has been a great home to many seabirds due to the high winds and the escape from land predators (Foster, 2026). In the 1920s–30s, Alcatraz was a highly secure prison with inadequate sewage systems, which scared away a lot of the bird populations that lived there. Thankfully, the Clean Air and Water Acts enacted in the 1960s–70s led to the closure of Alcatraz due to the poor sewage conditions, and now the small island is part of the Golden Gate Recreation Area. By the 1990s, herons, cormorants, gulls, and other seabirds called Alcatraz home again (Foster, 2026). This history of Alcatraz is a prime example of habitat loss and gain for birds, as well as an example of birds adapting to human conditions. Not only do birds live on Alcatraz, they also live on the Aquatic Park Pier.

After learning about Alcatraz from Corny, we walked along the pier. We observed a total of 27 taxa in Aquatic Cove and Fort Mason. One of the first birds we saw was a Brandt's Cormorant. There are 3 different species of cormorants in San Francisco: Brandt's, Double-crested, and Pelagic. It is common to see cormorants around the pier because Double-crested Cormorants nest under the Bay Bridge. As we watched the cormorants fish, Corny explained why cormorants are good swimmers and fishers. Cormorants are known to communally fish and they can dive 100 feet deep into the water (Foster, 2026).

As we walked up to the Fort Mason Community Garden, we learned about different fish-eating patterns. We saw gulls and learned that, unlike cormorants who communally fish, gulls fight for food because of how their mothers feed them as chicks. Seabirds are unique because they hunt by smell. While this is a useful skill, it does not protect them from anthropogenic pollution. While observing albatross, Corny explained that they eat algae from the bay that is infected with plastic and feed it to their chicks, often leading to their death (Foster, 2026). Our field trip allowed us to bring our education outside of the classroom and get a better understanding of the common birds around San Francisco Bay. In order to protect bird populations, we need birders like Corny for tracking data, and we need to monitor the pollution in San Francisco Bay to minimize the pollution birds in the bay come in contact with.

Conclusion

This paper discussed the decline of North American avifauna, the effect of pollutants on birds, and our observations at San Francisco Aquatic Park Cove. The overall decline in bird populations is a big concern due to their necessity in ecosystems and their role as an indicator of overall ecosystem health. Legislation targeting both habitat protection and mercury monitoring is essential to protecting San Francisco Bay birds and the broader ecosystem. In order to protect these birds, we need continued monitoring of bay pollution, strong conservation legislation, and birders like Corny Foster who make population tracking possible in the first place.

Work Cited

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